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1.3 Accuracy, Rounding and Estimation

OCR GCSE Mathematics (J560) — Component 01 (Paper 1, Non-Calculator), Component 02 (Paper 2, Calculator), Component 03 (Paper 3, Calculator)

1.3.1 Rounding

Rounding replaces a number with an approximate value that is simpler to work with. The three main types of rounding you need at GCSE are: to a given number of **decimal places (d.p.)**, to a given number of **significant figures (s.f.)**, and to the **nearest whole number, 10, 100** etc.

The Rounding Rule

Look at the digit *immediately to the right* of the position you are rounding to:

- If it is **5 or more** → round **up**
- If it is **4 or less** → round **down** (keep the digit unchanged)

Rounding to Decimal Places

Count from the decimal point. The digit in the required d.p. position is the *last digit you keep*. Look at the next digit to decide whether to round up or down.

Number	1 d.p.	2 d.p.	3 d.p.
3.7461	3.7	3.75	3.746
8.2950	8.3	8.30	8.295
0.00483	0.0	0.00	0.005

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12.9995	13.0	13.00	13.000

Exam Tip: When a question asks for 2 d.p., if the last digit rounds to zero you must write it: 8.30, not 8.3. The zero is significant — it shows the answer is given to 2 d.p.

Rounding to Significant Figures

Significant figures are counted from the **first non-zero digit**, left to right. Leading zeros are not significant; trailing zeros after a decimal point are significant.

Rules for counting significant figures:

- Non-zero digits are always significant: 3.14 has 3 s.f.
- Zeros between non-zero digits are significant: 2007 has 4 s.f.
- Leading zeros (before the first non-zero digit) are NOT significant: 0.0045 has 2 s.f.
- Trailing zeros after the decimal point ARE significant: 3.40 has 3 s.f.
- Trailing zeros in a whole number are ambiguous without a decimal point: 200 might have 1, 2, or 3 s.f.

Significant Figures: 0.004705 has 4 significant figures

In 0.004705, the leading zeros are not significant. The significant figures are 4, 7, 0, 5 — giving 4 s.f.

Number	1 s.f.	2 s.f.	3 s.f.
4,836	5,000	4,800	4,840
0.07382	0.07	0.074	0.0738
9.995	10	10.0	10.0

130,500	100,000	130,000	131,000

Worked Example:

Round 0.007358 to 2 significant figures.

First significant figure: 7 (the first non-zero digit)

Second significant figure: 3

Next digit: 5 → round up

Answer: 0.0074

Rounding to the Nearest 10, 100, 1000

Same rule: look at the digit to the right of the target place. If it is 5 or more, round up.

Worked Example:

Round 4,653 to the nearest 100.

Hundreds digit is 6. The next digit is 5 → round up.

Answer: 4,700

Exam Tip: When rounding to significant figures, use zeros as placeholders to maintain the correct magnitude. 4,836 rounded to 1 s.f. is 5,000 — not just 5.

1.3.2 Estimation

Estimation is used to check whether an exact answer is reasonable, or to get an approximate answer quickly — especially on Paper 1 (non-calculator). The standard approach is to round every number to **1 significant figure** before calculating.

To estimate a calculation:

1. Round every number in the calculation to 1 significant figure.
2. Carry out the simplified calculation.
3. State that your answer is an approximation (use the $\approx$ symbol).

Worked Example:

Estimate $\dfrac{38.4 \times 5.9}{0.196}$

Round each number to 1 s.f.: $38.4 \approx 40;\ 5.9 \approx 6;\ 0.196 \approx 0.2$

$$\frac{40 \times 6}{0.2} = \frac{240}{0.2} = 1200$$

The actual answer is ≈ 1155. The estimate 1200 is close — confirming no major error.

Worked Example:

Estimate $\sqrt{97} \times 4.8$

$\sqrt{97} \approx \sqrt{100} = 10;\ 4.8 \approx 5$

$$10 \times 5 = 50$$

Worked Example:

A cinema sells 287 tickets at £8.70 each. Estimate the total revenue.

$287 \approx 300;\ 8.70 \approx 9$

$$300 \times 9 = £2{,}700$$

When Not to Round to 1 s.f. Exactly

Sometimes rounding to 1 s.f. makes numbers awkward. If a number like 4.9 rounds to 5 but is close to 5 anyway, or if dividing by 0.2 is much harder than you'd like, you may choose convenient nearby

numbers instead — the important thing is that your working is clear and your estimate is reasonable.

> **Exam Tip:** On the exam, show your rounded values clearly. Write something like "$38.4 \approx 40$, $5.9 \approx 6$" before calculating. This earns method marks even if you make an arithmetic slip.

1.3.3 Error Intervals and Bounds

When a measurement is given to a particular degree of accuracy, the *true value* lies in a range. This range is described using an **error interval** or by giving the **lower** and **upper bounds**.

- **Lower bound:** the smallest value that would round to the given measurement.
- **Upper bound:** the largest value that would round to the given measurement.
- The upper bound is exclusive — it is the value that would just round *up* to the next value. Conventionally written with a strict inequality $<$.

Error interval notation: $\text{lower bound} \leq x < \text{upper bound}$

Finding Bounds When Rounding to a Unit

The key rule: the error is **half the unit of rounding**. Subtract and add this from the given value to find the bounds.

Rounded to	Unit of rounding	Half unit (error)	Example	Error interval
Nearest whole number	1	0.5	$x = 7$	$6.5 \leq x < 7.5$
Nearest 10	10	5	$x = 50$	$45 \leq x < 55$
Nearest 100	100	50	$x = 300$	$250 \leq x < 350$
1 decimal place	0.1	0.05	$x = 7.3$	$7.25 \leq x < 7.35$

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2 decimal places	0.01	0.005	$x = 4.62$	$4.615 \leq x < 4.625$
1 significant figure	varies	half the place value	$x = 400$ (1 s.f.)	$350 \leq x < 450$

Number Line Diagram for Bounds

Error interval for a length $l = 7.3$ cm (rounded to 1 d.p.). The solid dot shows the lower bound is included; the open circle shows the upper bound is excluded.

Worked Example:

A length is measured as 7.3 cm to 1 decimal place. Write the error interval for the length l.

Unit of rounding: 0.1 cm → half unit = 0.05 cm

Lower bound: $7.3 - 0.05 = 7.25$ cm

Upper bound: $7.3 + 0.05 = 7.35$ cm

$$7.25 \leq l < 7.35$$

Worked Example:

A crowd is estimated as 24,000 people, rounded to the nearest 1,000. Write the error interval.

Unit = 1,000; half unit = 500

$$23{,}500 \leq n < 24{,}500$$

Worked Example:

A number x is given as 0.06 to 1 significant figure. Find the error interval.

The significant figure is in the hundredths column (0.06); unit = 0.01; half unit = 0.005

$$0.055 \leq x < 0.065$$

Calculations with Bounds (Maximum and Minimum Values)

When a calculation involves multiple rounded values, use the appropriate bounds to find the maximum or minimum result.

- **Maximum result of $a + b$:** use upper bounds of both
- **Minimum result of $a + b$:** use lower bounds of both
- **Maximum result of $a - b$:** use upper bound of a and lower bound of b
- **Minimum result of $a - b$:** use lower bound of a and upper bound of b
- **Maximum result of $a \times b$:** use upper bounds of both (when both positive)
- **Maximum result of $a \div b$:** use upper bound of a and lower bound of b (when both positive)
- **Minimum result of $a \div b$:** use lower bound of a and upper bound of b (when both positive)

Worked Example:

A rectangle has length $l = 8.4$ cm and width $w = 3.7$ cm, both measured to 1 d.p.

Find: (a) the maximum area (b) the minimum perimeter.

Bounds for l: $8.35 \leq l < 8.45$

Bounds for w: $3.65 \leq w < 3.75$

(a) Maximum area uses upper bounds: $8.45 \times 3.75 = 31.6875 \text{ cm}^2$

(b) Minimum perimeter uses lower bounds: $2(8.35 + 3.65) = 2 \times 12 = 24 \text{ cm}$

Worked Example:

$p = 5.8$ cm and $q = 2.3$ cm, both rounded to 1 d.p. Find the minimum value of $\frac{p}{q}$, giving your answer to 3 s.f.

Minimum of $p \div q$: use lower bound of p and upper bound of q

Lower bound of $p = 5.75$; upper bound of $q = 2.35$

$$\frac{5.75}{2.35} = 2.446... \approx 2.45 \text{ (3 s.f.)}$$

Worked Example:
A car travels 180 miles (to the nearest 10) in 3.2 hours (to 1 d.p.). Find the maximum possible average speed.

Maximum speed = maximum distance ÷ minimum time

Upper bound of distance: 185 miles; lower bound of time: 3.15 hours

$$\text{Max speed} = \frac{185}{3.15} = 58.73... \approx 58.7 \text{ mph (3 s.f.)}$$

Exam Tip: The upper bound is always *strictly less than* (<), not ≤. A length of 7.35 would round to 7.4, not 7.3. Write error intervals as $7.25 \leq l < 7.35$, not $7.25 \leq l \leq 7.35$.

Exam Tip: When finding the maximum value of a division $a \div b$, you want the biggest top and the smallest bottom. When finding the minimum, you want the smallest top and the biggest bottom. Always think about which combination makes the result biggest or smallest.

1.3.4 Truncation vs Rounding

There are two distinct ways a number can be approximated to a given number of digits:

- **Rounding**: look at the next digit and round up if it is 5 or more, otherwise keep the last digit unchanged.
- **Truncation**: simply remove (chop off) all digits beyond the required position — regardless of what they are. There is no rounding up.

Number	Rounded to 2 d.p.	Truncated to 2 d.p.
3.7461	3.75	3.74
8.2950	8.30	8.29
1.9999	2.00	1.99
0.00583	0.01	0.00

Error Intervals for Truncated Values

When a value x is truncated to n decimal places, the error interval is **one-sided**: the true value is at least as large as the truncated value, but strictly less than one unit of the last place above it.

If x is truncated to 1 decimal place to give 4.7:

$$4.7 \leq x < 4.8$$

The lower bound *equals* the truncated value (no rounding means the value could be exactly 4.7).
The upper bound is the next value at that precision (4.8), exclusive.

Exam Tip: Truncation always gives a lower bound equal to the truncated value. Contrast with rounding, where the lower bound is below the given value (e.g. 4.7 rounded gives $4.65 \leq x < 4.75$, whereas 4.7 truncated gives $4.7 \leq x < 4.8$). If the question says "truncated", use the one-sided interval.

Worked Example:

A calculator displays 7.3 after truncating to 1 decimal place. Write the error interval.

Truncation: lower bound = 7.3 (exact), upper bound = 7.4 (exclusive)

$$7.3 \leq x < 7.4$$

Compare: if 7.3 had been *rounded* to 1 d.p., the interval would be $7.25 \leq x < 7.35$.

1.3.5 Percentage Error

The **percentage error** (or relative error) compares the absolute error to the true value. It is used to judge how significant an error is in context.

$$\text{Percentage error} = \frac{|\text{measured value} - \text{true value}|}{|\text{true value}|} \times 100\%$$

Or equivalently, using the maximum absolute error (half the unit of rounding):

$$\text{Maximum percentage error} = \frac{\text{half unit of rounding}}{\text{measured value}} \times 100\%$$

Worked Example:

A length is measured as 8 cm, rounded to the nearest centimetre. Find the maximum percentage error.

Half unit of rounding = 0.5 cm

$$\text{Max \% error} = \frac{0.5}{8} \times 100\% = 6.25\%$$

Worked Example:

A mass is given as 450 g, rounded to the nearest 10 g. Find the maximum percentage error to 3 s.f.

Half unit = 5 g

$$\text{Max \% error} = \frac{5}{450} \times 100\% = 1.11\% \ (3 \text{ s.f.})$$

Worked Example:

The true length of a piece of wood is 12.4 cm. A student measures it as 12.6 cm. Find the percentage error.

$$\text{Percentage error} = \frac{|12.6 - 12.4|}{12.4} \times 100\% = \frac{0.2}{12.4} \times 100\% \approx 1.61\%$$

Exam Tip: Percentage error questions often ask you to use the *measured* value in the denominator (as above), because the true value is unknown. Always use absolute values (ignore the sign of the error). A smaller percentage error means a more accurate measurement — this is more meaningful than the raw error alone.

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